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1. AXIOMS FOR A GROUP

To define a group, we begin with a set G equipped with a *composition law*

$$\cdot : G \times G \longrightarrow G, \quad (g, h) \longmapsto g \cdot h, \quad (1.1)$$

and a distinguished element $e \in G$, called the *identity*. Thus the basic data are written $(G, \cdot, e)$. We often suppress the dot and write gh in place of $g \cdot h$.

Definition 1.1 (Group). The data $(G, \cdot, e)$ form a *group* if the following three conditions hold.

(1) **Associativity.** For all $g, h, k \in G$,

$$g \cdot (h \cdot k) = (g \cdot h) \cdot k. \quad (1.2)$$

(2) **Identity.** For every $g \in G$,

$$e \cdot g = g \cdot e = g. \quad (1.3)$$

(3) **Inverses.** For every $g \in G$, there exists $h \in G$ such that

$$h \cdot g = e \quad \text{and} \quad g \cdot h = e. \quad (1.4)$$

The first equality says that h is a *left inverse* of g ; the second says that it is a *right inverse*. Thus an inverse is a two-sided inverse.

Associativity does *not* say that

$$g \cdot h = h \cdot g. \quad (1.5)$$

An operation satisfying this latter identity for all elements is called *commutative*.

Definition 1.2 (Abelian group). A group whose operation is commutative is called *abelian*.

It is also useful to observe the following simple result.

Lemma 1.1 (Uniqueness of inverses). *The element h in axiom (3) is unique.*

Proof. Suppose h_1 and h_2 both satisfy axiom (3) for g . Then

$$h_1 g = h_2 g = e. \quad (1.6)$$

Multiplying on the right by h_2 and using associativity gives

$$(h_1g)h_2 = (h_2g)h_2, \quad \text{hence} \quad h_1(gh_2) = h_2(gh_2). \quad (1.7)$$

Since $gh_2 = e$, it follows that $h_1 = h_2$. We write the unique inverse of g as g^{-1} . $\square$

2. ASSOCIATIVITY, CYCLIC GROUPS, AND ORDER

The associativity axiom allows us to write both

$$g \cdot (h \cdot k) \quad \text{and} \quad (g \cdot h) \cdot k \quad (2.1)$$

simply as

$$g \cdot h \cdot k. \quad (2.2)$$

More generally, given $g_1, \dots, g_n \in G$, repeated application of associativity allows us to define the element

$$g_1g_2 \cdots g_n \quad (2.3)$$

independently of how parentheses are inserted. Note again that we cannot change the order of the factors in the product unless the group is abelian.

For $x \in G$ and a positive integer n , we also write

$$x^n = \underbrace{x \cdot x \cdots x}_{n \text{ factors}}. \quad (2.4)$$

If $n = 0$, we set

$$x^0 = e, \quad (2.5)$$

where e is the identity of the group. If $n > 0$ and x^{-1} denotes the unique inverse of x , we set

$$x^{-n} = \underbrace{x^{-1} \cdot x^{-1} \cdots x^{-1}}_{n \text{ factors}}. \quad (2.6)$$

It then follows that

$$x^{-n} = (x^n)^{-1}. \quad (2.7)$$

Indeed,

$$(x^n)(x^{-n}) = \underbrace{x \cdot x \cdots x}_{n \text{ factors}} \underbrace{x^{-1} \cdot x^{-1} \cdots x^{-1}}_{n \text{ factors}} = e. \quad (2.8)$$

Definition 2.1 (Order of an element). Let $x \in G$. The *order* of x , denoted $\text{ord}_G(x)$ or $\text{ord}(x)$, is the smallest positive integer n such that $x^n = e$. If no such n exists, we write $\text{ord}(x) = \infty$ and say that x has infinite order. In particular, $\text{ord}(e) = 1$.

Remark. Confusingly, we also say that if G is finite as a set then G has finite order. If G is infinite, we write $|G| = \infty$ and say that G has infinite order.

Here is a simple result.

Lemma 2.1. Let $(G, \cdot, e)$ be a group and let $x \in G$. Write

$$\langle x \rangle := \{x^n \mid n \in \mathbb{Z}\}. \quad (2.9)$$

Then $\langle x \rangle$ is a group of G , called the cyclic subgroup generated by x .

Proof. For all $m, n \in \mathbb{Z}$, associativity and the definitions of positive and negative powers give

$$x^n x^m = x^{n+m}, \quad (2.10)$$

so $\langle x \rangle$ is closed under the group operation. Associativity is inherited from G , the identity $x^0 = e$ belongs to $\langle x \rangle$, and (2.7) shows that the inverse of x^n is x^{-n} . Thus $\langle x \rangle$ is a subgroup of G . $\square$

Here is a question to think about: certainly, if G has finite order, every element has finite order. But is the converse true? That is, if every element of G has finite order, must G be finite? Before we answer this question, let us turn to examples.

3. EXAMPLES

Example 3.1 (The integers under addition). The triple

$$(\mathbb{Z}, +, 0) \quad (3.1)$$

is a group. Its composition law is

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \quad (m, n) \longmapsto m + n. \quad (3.2)$$

The identity element is 0, and the inverse of m is $-m$.

Example 3.2 (The integers under multiplication: a non-example). The triple

$$(\mathbb{Z}, \times, 1) \quad (3.3)$$

is *not* a group, although it is a semigroup. Multiplication is associative and 1 is an identity, so axioms (1) and (2) hold. Axiom (3) fails because most integers do not have multiplicative inverses in $\mathbb{Z}$.

Example 3.3 (Integers modulo n). Let n be a positive integer. The set of residue classes modulo n is

$$\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}\}. \quad (3.4)$$

For example,

$$\begin{aligned} \bar{0} &= \{0, \pm n, \pm 2n, \pm 3n, \dots\}, \\ \bar{1} &= \{1, 1 \pm n, 1 \pm 2n, 1 \pm 3n, \dots\}. \end{aligned}$$

Every integer can be written $m = an + r$ with $0 \leq r < n$, and then $\bar{m} = \bar{r}$. Addition is defined by

$$\bar{m} + \bar{\ell} = \overline{m + \ell}. \quad (3.5)$$

Thus

$$(\mathbb{Z}/n\mathbb{Z}, +, \bar{0}) \quad (3.6)$$

is a group, called the *cyclic group of order n* .

In the preceding examples that are groups, the group operation is both associative and commutative.

Example 3.4 (The general linear group). The set

$$\mathrm{GL}_n(\mathbb{R}) = \{A : A \text{ is an invertible } n \times n \text{ matrix with entries in } \mathbb{R}\} \quad (3.7)$$

is a group under matrix multiplication. Its identity is the identity matrix

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}. \quad (3.8)$$

Thus the group is written $(\mathrm{GL}_n(\mathbb{R}), \cdot, I_n)$, and the composition law sends (A, B) to AB .

Example 3.5 (The general linear group). Consider now a subset of the previous example:

$$\mathrm{SL}_n(\mathbb{R}) = \{A : A \text{ is an invertible } n \times n \text{ matrix with entries in } \mathbb{R} \text{ and } \det(A) = 1\} \quad (3.9)$$

is a group under matrix multiplication. Its identity is the identity matrix

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}. \quad (3.10)$$

Why is it a group? What do we need to check ?

4. SUBGROUPS

Definition 4.1 (Subgroup). Let $(G, \cdot, e)$ be a group. A subset $H \subseteq G$ is called a *subgroup* of G if with the same composition as G and the same identity e , H itself satisfies the group axioms.

Now all subsets are groups! For example, the subset $\{0, 3\} \subset \mathbb{Z}$ is clearly not a group. However,

Proposition 4.1. Let $(G, \cdot, e)$ be a group and H a subset such that

- (1) $e \in H$.
- (2) H is closed under the group operation: if $h_1, h_2 \in H$, then

$$h_1 \cdot h_2 \in H. \quad (4.1)$$

- (3) H is closed under inverses: if $h \in H$, then

$$h^{-1} \in H. \quad (4.2)$$

Then H is subgroup of G .

Proof. The operation on a subgroup is the restriction of the operation on G . Associativity is therefore inherited from G , so it does not need to be checked separately. By condition (2) H is closed under this operation, and since e is an identity in G , it also satisfies the identity property of H . Similar for inverses. $\square$

Remark. Suppose $H \subseteq G$ is finite. If $e \in H$ and H is closed under the group operation, then H is automatically closed under inverses. Why? If $h \in H$ does not have an inverse in H , enumerate all elements in H as $\{h_1, \dots, h_n\}$ and consider the products: $\{hh_1, hh_2, \dots, hh_n\}$. I claim this set must be in bijection to H , since the map

$$H \rightarrow H, h_i \mapsto hh_i \quad (4.3)$$

is injective: if $hh_i = hh_j$ then multiplying both sides by the inverse of h in G , we obtain $h_i = h_j$. But an injective map of finite sets of the same cardinality is surjective, so there is some j such that $hh_j = e$.

Example 4.1 (The permutation group). The permutation group is

$$S_n = \{\text{all permutations of } \{1, 2, \dots, n\}\}. \quad (4.4)$$

Its group operation is composition of permutations. This is the example we will analyze in more detail next class.